

Cost analysis

Overdrive

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Summary

OverDrive – Cost analysis

1- General assumptions

2- Technical resources required

1. Application backend (REST API + WebSocket)
2. Relational database (PostgreSQL)
3. Caching and real-time management (Redis)
4. Object storage (assets and replays)
5. Time-series database (optional)
6. VR, mobile, and web clients
7. Development and collaboration tools

3- Estimated Monthly MVP Costs

1. Recurring infrastructure
2. Non-recurring costs and human costs

4- Conclusion – Cost Structure and Viability

1- General assumptions

The following analysis is built specifically for OverDrive, in its current configuration:

- Centralized backend (REST API + real-time WebSocket)
- PostgreSQL relational database
- Redis cache
- Object storage (assets, lightweight replays)
- VR client (Unity + MetaQuest)
- Companion mobile app
- Web configuration dashboard

The scope of the study corresponds to an advanced MVP (functional proof of concept), intended to :

- internal testing,
- demonstrations,
- a limited number of users (a few dozen).

Assumptions adopted

- Low to moderate load (one-off events, no mass production).
- Decent but not critical availability (tolerance for brief outages).
- Priority given to **cost control**, while maintaining a robust architecture for future scalability.
- Hosting on European cloud providers (e.g., Scaleway, OVHcloud).

2- Ressources techniques nécessaires

1- Application backend (REST API + WebSocket)

The OverDrive backend ensures :

- User account authentication and management
- Management of custom presets, layouts, and widgets
- Event and session catalog
- Real-time data streaming via WebSocket

For an MVP, a single instance (1–2 vCPUs, 2–4 GB RAM) is sufficient for :

- Host the backend (Go or Node/TypeScript),

- manage REST APIs,
- maintain active WebSocket connections.

Estimated cost (entry-level instance) :

5 à 15 € / month

2- Relational database (PostgreSQL)

PostgreSQL stores :

- Users and authentication sessions
- Custom presets and widgets
- Providers (championships), events, and sessions

A small managed plan (a few GB of storage, shared vCPU) is more than enough for an MVP.

Estimated cost (entry-level managed PostgreSQL) :

7 à 15 € / month

3- Caching and real-time management (Redis)

Redis is used for :

- The overall live status of a session
- The latest data per driver
- The high-frequency cache

Two strategies are possible :

1. Managed Redis (clean isolation)
2. Co-hosted Redis with the backend (cheaper, less isolated)

Coût estimé :

0 à 10 € / month

4- Object storage (assets and replays)

Use :

- Logos, images, UI elements
- Replays or lightweight exports

With an estimated volume between 5 and 20 GB, the costs remain very low.

Estimated cost :

1 à 3 € / month

5- Base time-series (optional)

For future advanced management of historical telemetry :

- InfluxDB Cloud
- TimescaleDB

As part of the MVP, this building block can :

- to be rejected,
- or used on a free / very basic plan.

Estimated cost :

0 à 10 € / month

6- VR, mobile and web clients

VR Client (Unity + Meta Quest)

- Unity: Free for students
- Main cost: Hardware

Meta Quest 3

approximately 500€ to 600€ (one-time purchase)

Companion mobile app

- Flutter/React Native development: free tools
- Minimal CI sufficient
- No significant direct costs

Dashboard web

- Very economical static hosting
- Or served by the existing backend

Estimated cost

0 à 3 € / month

7- Development and collaboration tools

Tool	Use	Cost
GitHub / GitLab	Source code + ticket management	Free (student plan)
Discord / collaborative tools	Team communication	Free
Monitoring (Sentry, logs)	Error tracking	Free (low volume)

3- Estimated monthly costs of the MVP

1- Recurring infrastructure

Ressource	Role	Assumption	Estimated cost / month
Backend server	API + WebSocket	1–2 vCPUs / 2–4 GB RAM	5–15 €
Managed PostgreSQL	Persistent data	Entry-level plan	7–15 €
Redis	Real-time cache	Managed or co-hosted	0–10 €
Object storage	Assets + replays	5–20 Go	1–3 €
Base time-series (optional)	Telemetry history	Basic/Free Plan	0–10 €
Web Dashboard	Static hosting	Basic	0–3 €

Overall monthly estimate

In an optimized MVP configuration: 20€ to 40€/month

This estimate depends :

- the chosen level of isolation (managed vs co-hosted),
- the performance safety margin,
- the actual volume of data.

2- Non-recurring costs and human costs

Matériel

- 1 to 2 Meta Quest headsets: 500€ to 600€ / unit
- Development machines (already owned by the team).

Human costs (order of economic magnitude)

If OverDrive were developed in a professional context:

- Real-time backend
- VR client
- Mobile app
- Web dashboard

The cost of a complete MVP would be estimated between 35,000€ and 75,000€ (depending on complexity, level of completion, daily rates)

In an academic context, these costs are internalized, but it is important to mention them to position OverDrive within a realistic economic framework.

4- Conclusion – Cost Structure and Viability

The architecture chosen for OverDrive allows for :

- Strong control over infrastructure costs
- Consistency with modern product standards
- Scalability towards a more ambitious version

In practice :

- **€20 to €40** per month is enough to host a credible MVP.
- A limited hardware investment (1–2 VR headsets) is sufficient to demonstrate the immersive value.
- The main cost remains human (design, development, and iteration time).

This cost structure confirms that OverDrive is **economically viable in the MVP** phase, while remaining aligned with a premium and scalable vision.